MILITARY SPECIFICATION SHEET

ELECTRON TUBE, CROSS FIELD AMPLIFIER

1/ TYPE 8934

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Broadband, pulsed, integral magnet, liquid cooled, 2,900 to 3,100 MHz, 60 kw rated peak power output

ABSOLUTE RATINGS: See note 27

Parameter:	epy	epx	ib	Ib	pi	Pi	tp	Du	VSWR	if	If	Ef	pd	F
Unit:	kv	kv	a	mAdc	kw	kW	us	---	---	a	A	V	kw	MHz
Maximum:	36	12	5	150	135	4	40	0.030	1.3:1	35	25	8	---	F3
Minimum:	---	---	3	---	---	---	---	0.006	---	---	---	---	1.6	F1
			(Note 1)				(Note 4)		(Note 3)		(Note 2)		(Note 5)	

ABSOLUTE RATINGS: See note 27

Parameter:	Waveguide pressurization	T(bushing)	Coolant inlet T(water)	Coolant inlet T(EG&W)	Coolant pressure (anode)	Coolant flow rate (anode)	Alt
Unit:	psia	°C	°C	°C	psia	gpm	ft
Maximum:	55	250	62	14	115	---	10,000
Minimum:	15	---	---	---	---	1.0	---
	(Note 6)	(Fig 1)	(Note 7)	(Note 7)		(Note 7)	

PHYSICAL CHARACTERISTICS:

Dimensions:	See figure 1	Cooling:	Forced liquid (see note 7)
Mounting position:	Cathode axis vertical	Marking:	See notes 1 and 2
Cathode connection:	See figure 1	Magnet isolation:	12 inches (see note 8)
Cooling connectors:	See figure 1	Handling:	See note 8
RF input and output couplings:	Mates with UG-54B/U type flange per MIL-F-3922/26	Weight:	55 lbs (max)

TEST CONDITIONS: See notes 9, 10, and 11

Parameter:	ib	Du	tp	VSWR	pd	Coolant inlet T(water)	Coolant flow rate (anode)	F
Unit:	a	---	us	---	kw	°C	gpm	MHz
Tolerance:	---	---	±0.5	Max	---	±15	---	---
Test condition 1:	Decal value (Note 1)	0.028	34.5 (Note 4)	1.1:1 (Note 3)	1.6 (Note 5)	20 (Note 7)	1.0 (Note 7)	F1,F2,F3 (Note 9)
Test condition 2:	Decal value (Note 1)	0.0068	34.5 (Note 4)	1.1:1 (Note 3)	1.6 (Note 5)	20 (Note 7)	1.0 (Note 7)	F1,F2,F3 (Note 9)

FREQUENCY	
F	MHz
1	2,900
2	3,000
3	3,100

GENERAL:

Qualifications - Required

1/ Replaces:- Raytheon type QKS-1629
ITT Gilfillan part number 145118-1

8934

FSC 5960

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METHOD	REQUIREMENT OR TEST	NOTES	TESTS	CONDITIONS	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 1	9,12	-					
4003	Pressurization (rf window)	6	-	55 psia (max)	---	---	---	---
4306	Pulse voltage	4	1,2	F1, F3	epy	24	27	kv
4250	Power output	11,13 11,13	1 2		Po Po	1,680 408	--- ---	W W
4279	Reverse directed power	11,14 11,14	1 2	VSWR = 1.3:1 (min) VSWR = 1.3:1 (min)	Pr Pr	--- ---	295 91	W W
---	Stability	11,15,16	1,2		MP	---	25	Pulses
---	Snap-on-stability	11,16,17	1		MP	---	25	Pulses
---	Stability (mode)	11,16,18	2	VSWR = 1.3:1 (min)	MP	---	25	Pulses
1261	Electrode voltage (getter)	-	-	I (getter) = 15 A; t = 300 (min)	E (getter)	1.0	5.0	V
	Quality conformance inspection, part 2							
4266	Direct-interelectrode capacitance	-	-		C	---	50	pF
4278	Phase shift	9,20	1,2		$\frac{\Delta\phi}{I\%\Delta ib}$	---	2.5	deg/a
---	Voltage variation	19	1		$\frac{\Delta V}{V}$	---	10	%
4153	Liquid cooling jacket leakage	6	-	p = 115 psi (min)	---	---	---	---
1155	Coolant-pressure drop versus coolant flow (anode)	7,21	-	Flow rate = 1 gpm	ΔP	1	5	psi
1105	Permanence of marking	-	-		---	---	---	---

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METHOD	REQUIREMENT OR TEST	NOTES	TESTS	CONDITIONS	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 3							
---	Life test	22,28	1	Group D	t	1,000	---	hrs
---	Life-test end points:							
4250	Power output	11,13	1		po	54	---	kw
	Stability	11,15, 16	1		MP	---	25	Pulses
	Periodic-check tests							
1032	Vibration, mechanical	23,24	-	No voltages	---	---	---	---
1042	Shock, specified pulse	24,25	-	Test conditions A and J	---	---	---	---
1011	Humidity	24	-		---	---	---	---
---	Temperature, altitude	24,26	-	Temperature, operating	T	-54	+60	°C
				Temperature, nonoperating	T	-65	+65	°C
				Altitude, operating	---	10,000	---	ft
				Altitude, nonoperating	---	30,000	---	ft

NOTES:

1. The optimum value of the peak operating current (Ib) shall be determined at F2. This value shall be designated the "decal value" and marked on each tube. The TUT shall be initially set to operate at the decal value; then the frequency changed to F1 and F3 without readjusting the high voltage.

2. The TUT filament current (If) shall be established as the lowest value which is consistent with normal operation. This value shall be designated the "decal value" and marked on each tube.

3. The system match looking away from the input and output flanges shall have a VSWR less than 1.5:1 for the frequency range of 2,600 to 3,200 MHz. The VSWR shall be less than 1.3:1 for the operating range of F1 to F3.

4. The pulse characteristics determined in accordance with method 4304 shall be as follows:

trv: 1.5 us (max) measured at 20 to 85 percent.
tfv: 5 us (max) measured at 20 to 85 percent.
tp: 34 us (min) measured at the 90 percent points.

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NOTES: -Continued

5. The following relationships shall exist between the TUT rf input and the rf output pulse:

   (a) t1 is the time interval between the leading edge of the rf input pulse and the leading edge of the TUT rf output which shall be between zero (minimum) and 1.0 us (maximum).
   (b) The TUT rf envelopes shall be positioned to be totally contained within the rf drive (pd) pulse.
   (c) Sufficient isolation from reverse directed power must exist between the rf input source and the TUT to maintain a high quality tube rf input drive signal.

6. The pressurization gas shall provide insulating properties which are the equivalent of oil-filtered and particle-filtered air dried to a dew point below 40°F. The oil trap may be activated alumina, the filter may be 20 to 30 micron size and the desiccant may be silica gel.

7. Demineralized water with a maximum inlet temperature of 62°C shall be used with, (1) specific resistivity maintained greater than 0.5 meg-cm, (2) oxygen content maintained less than 0.5 ppm, (3) an organic filter, and (4) a 3 to 4 micron filter at ambient temperatures in excess of +4°C. A 60 percent ethylene glycol and 40 percent distilled water solution, by volume, with a maximum inlet temperature of +14°C, with the same resistivity, oxygen content, and filters shall be used at ambient temperatures of +4°C and lower.

8. Extreme caution shall be used in the storage, handling, and installation of the tube to avoid imparting vibration or shock in excess of the values specified herein. To prevent demagnetization, ferromagnetic materials, energized magnets shall not be exposed to the tube magnet at a distance less than 12 inches.

9. The TUT shall be capable of satisfying all requirements specified herein at any frequency over the range of F1 thru F3, even though specific frequencies are specified for these tests.

10. Government owned equipment installed at the contractor's manufacturing facility shall be made available to perform all required tests specified herein. A transmitter test fixture with necessary microwave and modulator components, as specified in note 11, shall be used for these tests. Appropriate terminations shall be used where the TUT is required to operate into a VSWR of 1.3:1.

11. The TUT shall perform satisfactorily within the limits specified herein when tested in the figure 2 test circuit. The line-type modulator shall utilize a pulse-forming network whose impedance is within 10 percent of the value specified. (See note 10.)

12. Unless otherwise specified, the AQL for all tests listed under quality conformance inspection, part 1, shall be 1.0, inspection level II.

13. The TUT shall be capable of rated power output not less than the specified limit when tested at each 20 Mlz interval over the frequency range of F1 through F3.

14. The phase of the mismatched load shall be adjusted for maximum reverse directed power at the input of the TUT.

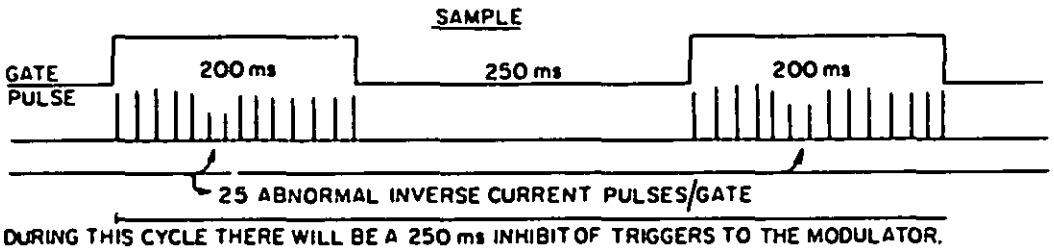
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NOTES: -Continued

15. The operating frequency shall be swept from F1 through F3 at a rate of 100 sec/sweep. The duration of this test shall not exceed 5 minutes. The TUT shall be rejected for:

    (a) Repeated waveguide breakdowns, or
    (b) Two successive inhibits of the modulator.

16. Stability shall be measured in terms of the average number of output pulses missing, normally expressed as a percentage of the number of input pulses applied during the period of observation. Output (rf) pulses are considered to be "missing" if, due to any causes, the average rf pulse energy, within a ±1 percent frequency range of the normal operating frequency, is less than 70 percent of the energy of a normal pulse. The number of missing pulses shall not exceed the specified limit, when measured in any two alternating 200 millisecond time periods. Observation shall be made at the load phase position corresponding to maximum instability.



17. The operating frequency shall be swept from F1 through F3 at a rate of 100 ms/sweep. The anode current shall be snapped on at that level required for rated power output specified herein. The TUT shall be rejected for:

    (a) Repeated waveguide breakdown.
    (b) Two successive inhibits of the modulator.

18. The TUT shall be operated into a VSWR of 1.3:1 (minimum) varied over 360 degrees continuously or in steps of 90° ±10°.

19. The TUT voltage variation shall be less than 10 percent based on the voltage at F2, when the frequency is swept from F1 through F3. Relative voltage indications may be used for this test. The voltage variation shall be computed as follows:

$$\frac{VF3 - VF1}{VF2} < 10\%$$

20. The anode current shall be varied about the decal value. The incremental phase shift ($\Delta\phi$) value shall be computed from the difference between the phase shifts through the TUT at the two anode current values divided by the total peak current change.

    Definition: $\dfrac{\Delta\phi}{\Delta\dfrac{ib \times 100}{ib}}$ = Incremental phase shift for 1 percent change in anode current.

21. The pressure drop (ΔP) shall be measured between the inlet and outlet connectors. There shall be no evidence of leaks.

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NOTES: -Continued

22. A life-test cycle shall consist of the following schedule:

Condition	Duration
High voltage "on"	7.5 hours
High voltage "off"	0.5 hours

The frequency shall be varied from F1 through F3 and repeated after each 100 hours operating time.

23. The TUT shall be mounted with the cathode axis vertical and vibrated as follows:

Exploratory:	Frequency	Amplitude (peak to peak)
	5 to 15 Hz	0.075 ±0.012 (0.150 total displacement)
	16 to 25 Hz	0.050 ±0.008 (0.100 total displacement)
	25 to 33 Hz	0.025 ±0.004 (0.050)total displacement)

Variable: As specified in the exploratory test except that the dwell time at each frequency with observed resonances shall be 5 minutes (approximately).

Endurance: As specified in the variable test except that the dwell time shall be 2 hours (minimum).

24. After the completion of this test, the TUT shall exhibit no mechanical failures and shall satisfy all requirements of the pulse voltage, power output, and stability tests of quality conformance inspection, part 1.

25. The TUT shall be mounted with the cathode axis vertical and shocked in accordance with condition A in a direction along the cathode axis. The TUT will then be shocked in accordance with condition J (except 25 G) along the other two mutually perpendicular axes.

26. The TUT shall be paced in the test chamber in a manner similar to that in which it will be used in service, and such electrical and mechanical connections and instrumentations as may be necessary shall be completed. When changing chamber conditions from those required for one step to those required for any other step, the rates of temperature and pressure changes shall be as high as chamber construction permits, except that these rates shall not exceed 1°C (1.8°F) per second and 0.5 inches of mercury per second respectively. The change from room temperature to -62°C (-80°F) shall not exceed 4 hours. The temperature altitude conditions shall be as specified in the following steps:

Step	Chamber temperature a/	Pressure/altitude (simulated)	Operating time b/ (minutes)
1	-65°C (-85°F) (1 hr minimum)	Atmospheric	---
2	-54°C (-65°F)	Atmospheric	60
3	-54°C (-65°F)	30,000 ft	60
4	+85°C (185°F)	Atmospheric	120
5	+30°C (86°F)	Atmospheric	60
6	+55°C (130°F)	Atmospheric	4 cycles c/
7	+30°C (86°F)	Atmospheric	60
8	+66°C (151°F)	Atmospheric	15
9	+30°C (86°F)	30,000 ft	120
10	+59°C (138°F)	30,000 ft	30

a/ Thermally stabilize the TUT under nonoperating conditions.
b/ Monitor the power output with the TUT operating under conditions of test 1.
c/ Each cycle shall consist of 30 minutes "on" (operating) and 15 minutes "off".

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NOTES: -Continued

27. In addition to the MIL-E-1 symbols, the following shall apply:

EG & W	Ethylene glycol and water
gpm	Gallons per minute
P	Pressure
ΔP	Pressure drop
Pr	Reverse power
Δϕ	Phase change

28. Service-life guarantee may be acceptable if specifically authorized by contract.

Custodian:
Navy - EC

Review activity:
DSA - ES

User activities:
Navy - AS, CG, MC, OS, SH

Preparing activity: Navy - EC

Agent: DSA - ES

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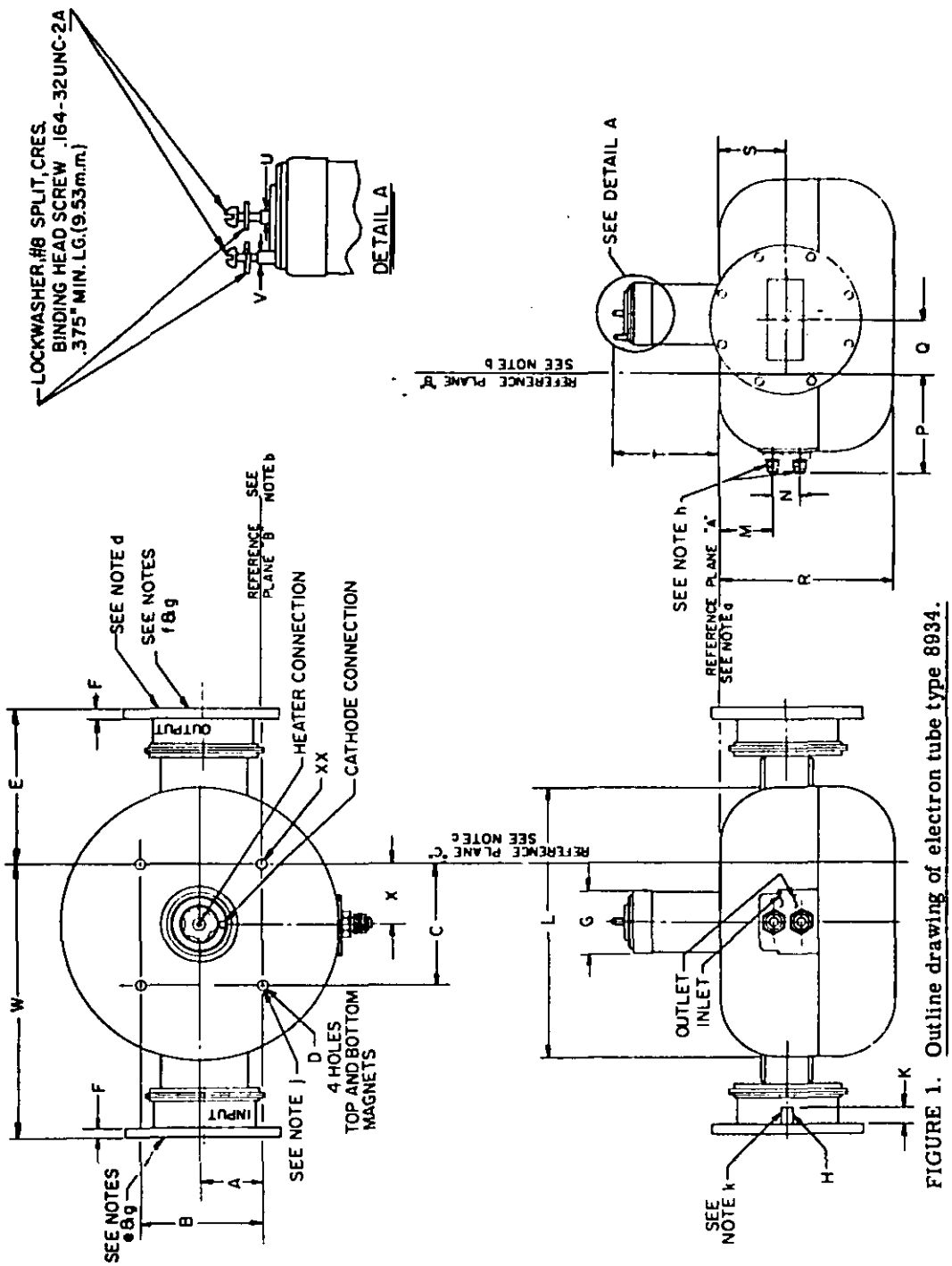


FIGURE 1. Outline drawing of electron tube type 8934.

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LTR	Dimensions in inches with metric equivalents (mm) in parentheses.				NOTES
	MINIMUM		MAXIMUM		
	Quality conformance inspection, part 1				
D	.300	(7.620)	.340	(8.636)	
E	5.250	(133.35)	5.550	(140.97)	
F	.290	(7.37)	.310	(7.87)	
G	---	---	2.200	(55.88)	
L	---	---	10.470	(265.94)	
M	1.715	(43.56)	2.015	(51.18)	
N	.980	(24.89)	1.020	(25.91)	
Q	1.788	(45.42)	2.188	(55.58)	
R	6.225	(158.12)	6.325	(160.66)	
T	---	---	4.000	(101.60)	
U	.240	(6.10)	---	---	
V	.240	(6.10)	---	---	
W	14.990	(380.75)	15.110	(383.79)	
Quality conformance inspection, part 2					
A	2.000	(50.80)	2.250	(57.15)	
B	4.210	(106.94)	4.290	(108.97)	
C	4.210	(106.94)	4.290	(108.97)	
H	.2385	(6.05)	DIA	---	k
K	.500	(12.70)	.760	(19.30)	
P	---	---	4.062	(103.17)	
S	2.360	(59.94)	2.610	(66.29)	
X	2.000	(50.80)	2.250	(57.15)	

NOTES:

a. Reference plane "A" is defined as a plane which passes along the face of the magnet as shown.
b. Reference plane "B" is defined as a plane perpendicular to plane "A" and passing through the axis of the holes at plane "A" as shown.
c. Reference plane "C" is defined as a plane mutually perpendicular to plane "A" and "B" passing through the axis of hole "XX" at plane "A" as shown.
d. Waveguide flanges to be parallel to ref. plane "C" within .050 (1.270 mm).
e. RF input.
f. RF output.
g. Flange to mate with UG54B/U choke flange.
h. Anode coolant connections fittings shall be AN815-6 or equivalent per MIL-F-5509.
j. The four holes shall be capable of simultaneously accepting four .250 inch (6.35 mm) diameter by 6.50 inch (165.10 mm) minimum length rods.
k. Ground connection. 250-20 UNC-2A on flange bolt circle.

FIGURE 1. Outline drawing of electron tube type 8934 - Continued.

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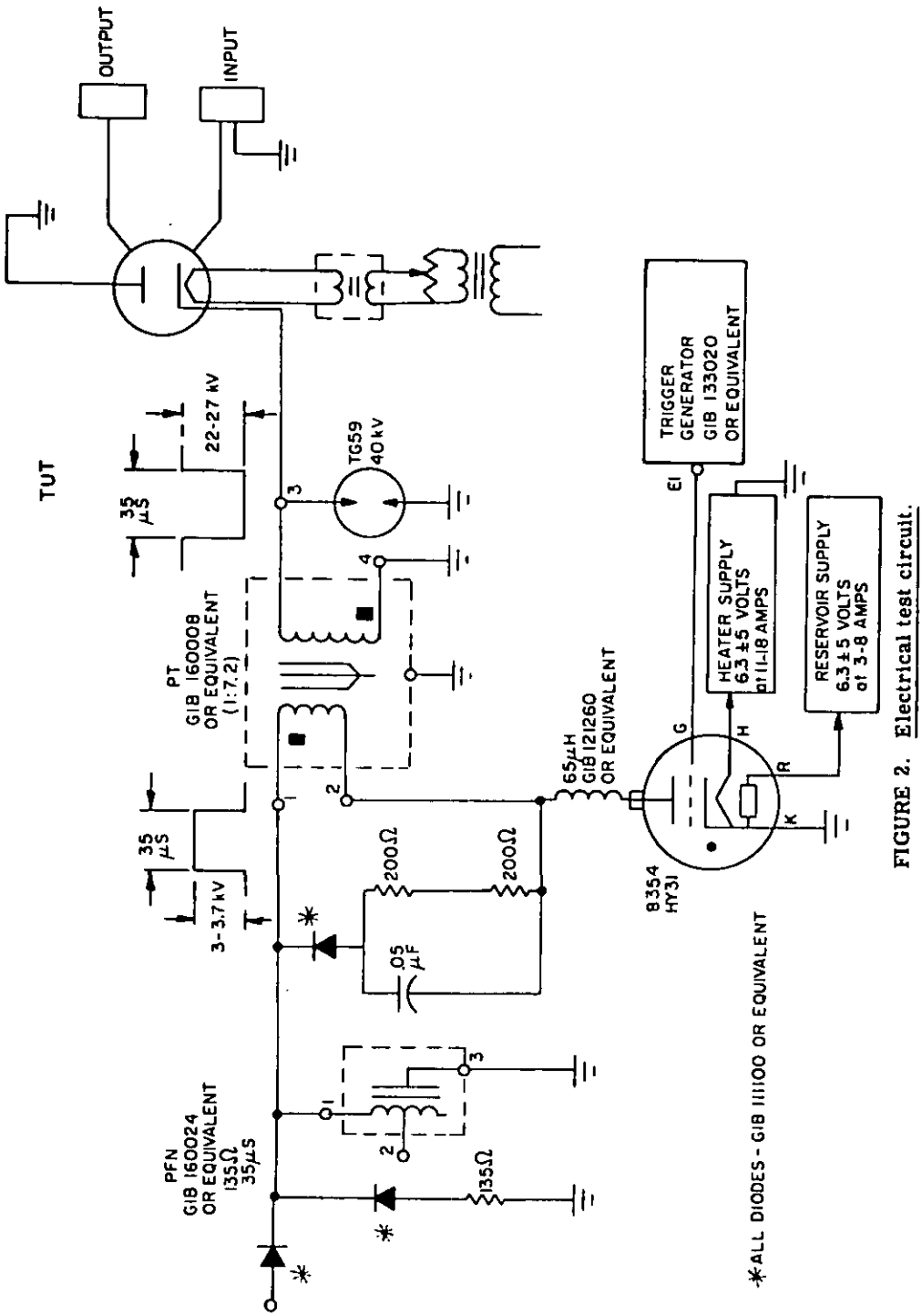


FIGURE 2. Electrical test circuit.

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